

# Splunk Search — Beginner Cheat-Sheet

Quick reference companion to 'Splunk Search for Beginners'

SPL | filter early, filter cheap

## ANATOMY OF A SEARCH

Read left to right; the pipe | feeds each command's output into the next.

```
index=web sourcetype=access_combined status>=500
| stats count by host
| sort -count
```

**base search** (retrieve) → **transform** → **format**. Filter early, filter cheap.

## METADATA FIELDS (put these first)

|            |                                             |
|------------|---------------------------------------------|
| index      | Where stored (bucket of data on disk)       |
| host       | Which machine produced it                   |
| source     | Which file/input it came from               |
| sourcetype | What format it is (drives field extraction) |

Known before raw text is read → cheapest, fastest filter. No index=? You only search your role's defaults.

## SYNTAX RULES & GOTCHAS

- Multiple terms are AND by default: a b = a AND b.
- AND / OR / NOT must be UPPERCASE.**
- Precedence: () > NOT > OR > AND. Use parens when unsure.
- Wildcard \*: fail\* matches fail, failed, failure. **Avoid leading \*term** (slow).
- Quotes for phrases/spaces: "login failed".
- Keyword search is case-INSENSITIVE; field *values* often are case-SENSITIVE.
- error (keyword, anywhere in text) vs status=error (exact field value, faster).

## TIME RANGES

Set via the time picker, or inline with earliest / latest:

|                |                               |
|----------------|-------------------------------|
| earliest=-24h  | Last 24 hours                 |
| earliest=-7d@d | Last 7 whole days to midnight |
| latest=@d      |                               |
| earliest=-15m  | Last 15 minutes               |

Units: s m h d w mon y. @ snaps to a boundary (@d = start of today).

## SEARCH MODES

|         |                                 |
|---------|---------------------------------|
| Fast    | Only fields you name. Quickest. |
| Smart   | Default. Balances the two.      |
| Verbose | Every possible field. Slowest.  |

Missing fields in the sidebar? You may be in **Fast** mode.

## COMMON COMMANDS

| Command     | What it does                                   |
|-------------|------------------------------------------------|
| stats       | Aggregate (count/sum/avg), optionally by group |
| top / rare  | Most / least common values of a field          |
| timechart   | Aggregate over time (trends)                   |
| chart       | Aggregate across two dimensions                |
| table       | Pick columns to display                        |
| fields      | Keep/drop fields (speeds searches)             |
| sort        | Order results (- = descending)                 |
| dedup       | Remove duplicate values of a field             |
| eval        | Create or modify a field                       |
| where       | Filter using an expression                     |
| rename      | Rename a field                                 |
| head / tail | Keep first / last N results                    |

## STATS FUNCTIONS

| Function       | Returns                                 |
|----------------|-----------------------------------------|
| count          | Number of events (only one usable bare) |
| dc(field)      | Distinct/unique count                   |
| sum(field)     | Total of a numeric field                |
| avg(field)     | Mean                                    |
| min/max(field) | Smallest / largest value                |
| values(field)  | List of UNIQUE values                   |
| list(field)    | List of ALL values (with dups)          |

Rename outputs with AS: dc(clientip) AS visitors. After stats, raw events are gone—use the new fields downstream.

## STATS vs CHART vs TIMECHART

| Command                                  | Output shape                     |
|------------------------------------------|----------------------------------|
| stats                                    | Flat table, one row per group    |
| chart                                    | Matrix: ... over X by Y          |
| timechart                                | Time is the grouping (line/area) |
| stats count by X = "how many of each X?" |                                  |

## NO RESULTS? CHECK IN ORDER

- **Time range** — #1 cause. Widen to All time to confirm.
- **Wrong index** / one your role can't see. Try naming index= or index=\*
- **Value case / exact match** — status=0K ≠ ok.
- **Wrong field name** or field not extracted. Check the sidebar.
- **Typos / unbalanced** ( or " .

Tip: strip back to the base search, then add one filter at a time until results vanish.

## MAKE SEARCHES FASTER

- Narrow the **time range** first.
- Lead with index / sourcetype / host.
- Put most specific filters at the front.
- Avoid leading wildcards \*term.
- Trim early: | fields a, b.
- Prefer stats over pulling raw events.